**The Evolution of a Statewide Model for New Jersey**

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## INTRODUCTION

The state of New Jersey has twenty one counties, and is located between two major cities, Philadelphia and New York City. There are three Metropolitan Planning Organizations (MPOs) in the state: the North Jersey Transportation Planning Agency (NJTPA), the Delaware Valley Regional Planning Commission (DVRPC), and the South Jersey Transportation Planning Organization (SJTPO). The NJTPA region consists of the 13 northern New Jersey counties, and is a bridge crossing away from New York City. It is heavily influenced by the City. The DVPRC region consists of four New Jersey counties and five Pennsylvania counties including the City of Philadelphia. The SJTPO region covers the remaining four counties and includes Atlantic City. This region includes most of the New Jersey shore/beach area, casinos and summer recreational markets.

The three uniquely different MPO regions are represented by three unique transportation demand models. There are three transportation demand models corresponding to each of the MPOs. These models were originally developed to be used for regional transportation planning by the New Jersey Department of Transportation (NJDOT) and the MPOs. Later, with the passing of the 1990 Clean Air Act and ISTEA, the expectations and uses for these models changed significantly. They would now be used for macro scale and micro scale types of analyses. New uses included support of the Transportation Improvement Programs (TIP), State Implementation Plans (SIP), Plan Conformity, TIP/SIP Conformity, air quality budgets, and others items. These additional demands on the models would result in greater scrutiny of the models' results and assumptions.

## THE NORTH JERSEY REGIONAL TRANSPORTATION MODEL (NJRTM)

The NJRTM was developed using the TRANPLAN transportation planning software package. The NJRTM includes the 13 North Jersey counties and contains over 1450 census tract based zones. New York City, New York State and Eastern Pennsylvania are external stations in the model and are not included as part of the zonal system. This could be considered as a weakness, but in the development of the model there were software limitations on the maximum number of zones that could be modeled. Consequently, the decision was made to stop the model at the Hudson River and the Delaware River. The NJRTM includes the traditional four step model process of:

1. trip generation
2. trip distribution
3. mode choice
4. highway & transit assignments

Because of the close proximity to New York City, this region has a significant share of transit trips. The transit trips that remain in this 13 county region are estimated by this model, however, transit trips between New York City and Jersey are estimated by a New Jersey Transit-Hudson River Waterfront model and imported into the NJRTM. The NJ Transit model was developed by NJ Transit for their planning and operation. Their model uses MINUTP and has a more complex mode choice model and more network details in the urban areas and CBDs. It also included Manhattan, which is why the New York trips are imported into the NJRTM.

The NJRTM was initiated in 1986, and calibrated based on travel characteristics and patterns from the 1986/87 Home Interview Survey (HIS), and the 1980 census. It has four trip purposes, HBWork, HBShop, HBOther, and NHB. The trips were stratified this way based on the results from the HIS. This model was enhanced during the mid 1990s to include the 1990 census results, time-of-day analysis, accessibility function and feedback process. These enhancements allows the model to meet the requirements identified in the Best Practice Guidelines.

**THE DVRPC MODEL**

The DVRPC model is a four step model of four New Jersey counties, Mercer, Burlington, Camden, and Gloucester, along with five Pennsylvania counties. After years of running this model in UTPS, it now operates in the TRANPLAN. The zonal system for this model is based on census tracts including the City of Philadelphia and surrounding counties. There are 1395 internal zones, with over 350 New Jersey zones. The Delaware River crossing trips between Pennsylvania and Jersey are internal trips in the model. The DVRPC model has three trip purposes, HBWork, HBNWork, and NHB. It also includes six area types, CBD, Fringe CBD, Urban, Suburban, Rural and Open Rural.

The early foundation of this model is based on the 1960 travel survey data. This model has been updated based on the 1987 HIS and the 1990 CTPP Journey-to-Work data. Additional O-D data and transit ridership data was also utilized. Because transit trips into and around the City of Philadelphia are an important component of this model, additional network was required and in the CBD the zone system was disaggregated to the tract level. A study to evaluate this model was recently completed, and identified some enhancements that could be made to the various models.

**THE SJTPO MODEL**

The primary counties included in this model are Atlantic, Cape May, Cumberland, and Salem. These four counties makeup the SJTPO Region. Salem and Gloucester are not in the region but are included in the model as secondary counties. This was done to better analyze South Jersey issues. The components of this model are trip tables and an assignment tool. The trip table is based on trip generation and distribution from an 1980s model and extensive roadside surveys which were conducted in 1989. The model simulates a summer Friday condition with five trip purposes; work, casino work, casino visit, beach visit, other. Most of the congestion problems in this MPO are seasonal related (mainly summer months) and this model has been effective for more than ten years. Transit usage is not significant in this region, and until recently there existed no real need to use the model for transit planning. However, this is a rapidly changing area and the model is no longer acceptable in developing future transportation demand forecast.

## THE STATEWIDE MODEL

During the early 1990s as New Jersey made plans to complete the Interstate system it became apparent that a tool was needed to analyze projects that crossed the above model boundaries, and state boundaries. A tool was necessary to evaluate the impact of a project in one region on another region. To accomplish this would require building a statewide model, including the 21 counties and the three regional models. It was also decided that the area surrounding New Jersey should be included in the new statewide model. Instead of building a new model to accomplish the objectives, it was decided to try and combine the five existing regional models that cover the state and areas of interest:

1. North Jersey Regional Transportation Model
2. South Jersey Regional Transportation Model
3. Delaware Valley Regional Planning Commission Model
4. Port Authority of NY/NJ Interstate Network Model
5. New Castle County Model from Delaware DOT

The benefits of this approach was that a new costly four step model would not have to be developed. This would save time and money. This approach would also take advantage of the work already invested into the existing models.

An important component in the development of this new statewide model was the need and ability to analyze truck and goods movement within and through New Jersey. None of the existing models had the ability to separately evaluate trucks and goods movement, therefore development of a truck model as part of this statewide model was deemed necessary for this effort.

The above models are considered regional detail models, with sufficient detail and accuracy for incorporation into the new statewide model, and required no additional enhancements. It was decided that this new model would be formed through the merger of these five networks and trip tables. The zonal system and network was collapsed from 4023 zones and 71,502 links, to 2813 zones and 44,000 links. The New Jersey zones from the three New Jersey models remained unchanged at the census tract level, but several of the Pennsylvania and Delaware zones were aggregated.

The NJRTM has the largest portion of New Jersey zones of the three models. Therefore, it was decided that the NJRTM would be the foundation of the statewide model. The software package and the network format from the NJRTM was used. The other models networks were converted for consistency with the NJRTM. The networks were expanded to include some additional features, such as the coding of truck routes and nontruck routes, the coding of a separate toll structure for autos and trucks. Truck restriction for large trucks and no trucks were placed on the network. These changes were easily accomplished using dBASE IV and a spreadsheet software package. The external zones/cordons from the five models were connected to create a single network for the statewide model.

The auto trip tables from the five models would be combined to form one auto trip table for the statewide model. This was accomplished using a special FORTRAN technique called Trip Table Weaving. The first step in this process was to make the time of day, peak periods and daily trip tables consistent for the different models. The internal-internal trips from each model were incorporated into the statewide model unchanged. However, the Trip Table Weaving technique would weave the internal-external, external-internal, and external-external trips for two adjoining models. The external trip from one model would be distributed to the connecting zone based on that model's distribution of those trips. The final step in the process was to validate the final trip table and balance the external trips between the two adjoining models.

The truck trip tables were developed using a standard gravity model. It was based on commodity flows for the region. This commodity flow data was compiled to produce a county-to-county commodity trip table. This table was then converted to trucks based on truck inventory data and census data. The local delivery type truck trips were not included in the commodity data and had to estimated base on zonal employment and household data. It was also necessary to locate special truck generators. Examples of special truck generators are rail yards, ports, airports, landfills, truck terminals and warehouses. This detail produced a complete truck trip table capable of estimating regional and local truck traffic on network by truck size. Four truck classes were developed from these trip tables:

1. light trucks
2. medium trucks
3. heavy trucks
4. 102-inch (which are greater than the standard 96-inch width)

The first project to be analyzed using the new statewide model was the I-287 completion project. The limits of this project crossed all three MPO regions and the regions surrounding New Jersey. The impacts were primarily heavy trucks traveling long haul and through New Jersey. The model estimated the number of autos and trucks that would be diverted from the New Jersey Turnpike to I-287 to avoid tolls and New York City congestion. The results were later validated with traffic counts and truck O/D surveys at several locations.

This network merging and Trip Table Weaving techniques provided a cost effective method to create a statewide model. It utilized the best models presently available for the region. To build this model from the beginning would be costly and time consuming. This model has allowed the NJDOT and other outside agencies to evaluate significant projects that cross MPO boundaries, which otherwise could not be accomplished with any of the existing three MPO models.

## NEXT STEPS AND ENHANCEMENTS

As these regional transportation models continue to evolve, additional demands and expectations are placed on them. Previously these model were only required to predict motorized trips, but now transportation model developers are predicting non-motorized trips, and from the non-motorized trips, predicting bicycle and pedestrian usage. These regional models continue to be vital in air quality, but are now also being utilized in the Congestion Management Systems. They are used to test various transportation improvement strategies such as HOV lanes, congestion pricing, arterial signal systems and other intelligent transportation systems. Advances in personal computers have allowed many of these changes to occur; networks are expanding, zonal system are becoming more detailed, there are more feedback loops, and more iterations. To meet these demands New Jersey DOT, the MPOs and other transportation agencies must work together to maximize staff and resources. The agencies must continue to enhance the models and explore new ideas to ensure that the models remain up to date and capable of meeting the present and future demands.

In the North Jersey/New York Region, the New York Metropolitan Transportation Council (NYMTC), the MPO for the New York region, is developing a new transportation model for its region with many of the state-of-the-art practices. This region includes the North Jersey counties. The NYMTC model will become the third model developed for the North Jersey counties. Originally it was thought that this model could eliminate the need for the other two models, NJRTM and NJ Transit model. In the early stages of the NYMTC study it became apparent that the NJRTM and the NJ Transit model met certain needs that the NYMTC model could not meet. Since the NYMTC model would take years to complete, significant financial investments have gone into enhancing the other two models. However, as part of the NYMTC model development, an extensive HIS would be conducted to obtain data on trip generation, trip chaining, non-motorized trips and other essential items. New Jersey participated in this HIS to get additional survey samples for New Jersey counties, and this information would be the basis for the next revalidation of the NJRTM.

All three model will eventually be used to model regionally significant projects. Although they will be used by three different agencies with different missions, it will be important that each produces reasonable and consistent results for similar projects. To accomplish this similarity , each of the model should have similar networks and zonal systems. They will be based on the same HIS, and produce similar generation rates and equations. A important issue to be resolved will be the demographics and the selection of a land use model to ensure consistency. These issues are being discussed between the MPOs and various state agencies, because they impact air quality, emission budgets, statewide planning, and corridor planning. Both the NJTPA and the NYMTC are evaluating the various land use models.

Presently, the DVRPC model is being enhanced to include many of the state-of-the-art practices. Each of the components of this four step process will undergo a series of enhancements. Generation will be revised based on a HIS which is scheduled to begin in 1999. It will simulate non-motorized trips, bicycle and pedestrian travel. The mode choice model will become a nested logit modal split model and the transit and highway assignments will run for the peak and off peak periods separately. Another important addition to this model will be the incremental feedback procedure through the entire model process. This is a new feature in the TRANPLAN package. The DVRPC has received a grant to evaluate several of the land use packages for incorporation into this model.

A new four step model with mode choice and other state of the art practices is being validated for the SJTPO region. This new model will replace the old model. The first task in the development of this model was a beach survey at four different locations to understand the trip characteristics for beach travelers. With this data, a recreational trip generation model was developed, it includes 16 trip purposes (hbcasino access, nhbcasino visit, casino bus, hbevent, hbbeach access, hbbeach, hbboardwalk, hbshop, hbdine, hbo, nhbevent, nhbbeach, nhbboardwalk, nhbshop, nhbdine, and nhbo). A non-recreational trip generation model was also developed. It includes nine trip purposes(hbw, hbschool, hbshop, hbo, nhbw, nhbat-work, nhbo, heavy truck and commercial). This detail was provided to test various alternatives for Atlantic City and the many Jersey beaches. The SJTPO model zones are census tracts except for Atlantic City where a casino can be a single zone. Another important component of this model is the tabulation of Life Cycle into the production model. Life cycle is comprised of the following categories:

1. No retired people, no children age 18 or younger
2. No retired people, with children 18 or under
3. Retired people

The theory here is that the presence of children increases household trip making and the presence of retired people should reduce trip making. Life cycle was tested in other states and imported into this model. The high percentage of retirement communities on the Jersey shore, make this a valued enhancement.

Some of the techniques developed in the other models will be incorporated into the SJTPO model. The coverage of this model will be expanded by weaving trip tables and networks with DVRPC model. Other new items include land use accessibility, feedback and the temporal model. To improve the mode share model NJ Transit invested time and money into the mode choice process.

A unique aspect of this model is that it will be calibrated for an August Friday. But based on data collected for this study, factors were developed to adjust trip tables for any day of the week and any month of the year. This is an important component to this model. It allows the model to test the various strategies on the different days and the different seasonal characteristics.

The enhancements that will be performed on the NJRTM, DVRPC, and SJTPO models will be imported into the statewide model using the existing network merging and trip table weaving techniques. This will result in an improved statewide model. This work effort is not in the existing plans, but it will be a project in the future.

Over the last ten years, these models have undergone a series of enhancements costing millions of dollars. If the Transims traffic-simulation project is ever implemented in New Jersey it will cost millions more. In order to maintain these state-of-the-art models, and satisfy existing and future demands, New Jersey anticipates allocating significant resources and continuing this process for many years.